

Spherical Harmonics Coefficient Prediction for Gaussian Splat Coding

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ABSTRACT

3D Gaussian Splatting (3DGS) enables high-quality free-viewpoint rendering but suffers from large data sizes due to its high-dimensional attributes, particularly spherical harmonics (SH) coefficients. In MPEG Gaussian Splat Coding (GSC), both the video-based TMCV framework and fast-track V-PCC extensions are under developed; however, explicit tools that effectively exploit the structural correlation among SH coefficients remain limited. This paper proposes a prediction-based residual coding method that compresses prediction residuals as 2D attribute maps using HEVC by exploiting the correlation among SH coefficients within each color channel. The proposed method is implemented in both the Test Model Candidate Video (TMCV) and the V-PCC reference software (TMC2), and its performance is evaluated under the GSC common test conditions (CTC). For forward-facing sequences, the proposed method achieves average BD-rate reductions of 16.6% and 3.5% (in terms of RGB-PSNR) compared to the TMCV and V-PCC anchors, respectively. In contrast, average BD-rate increases are observed for object-centric sequences, motivating future work on adaptive application of the proposed SH coefficient prediction method depending on sequence characteristics.

Index terms: 3D Gaussian Splatting (3DGS), Gaussian Splat Coding (GSC), TMCV, V-PCC, Spherical Harmonics (SH) coefficients.

1. INTRODUCTION

3D Gaussian Splatting (3DGS) [1] has attracted significant attention as a 3D scene representation approach that models a scene using hundreds of thousands of 3D Gaussian primitives and enables high-quality, efficient free-viewpoint rendering. A 3D Gaussian primitive is represented by a 59-dimensional attribute vector, including position, rotation, scale, opacity, and spherical harmonics (SH) coefficients that characterize color and directional illumination. Since these attributes must be stored for a large number of primitives, the resulting 3DGS model size becomes

very large, making efficient compression essential for practical storage and transmission.

Accordingly, MPEG has been promoting standardization activities under Gaussian Splat Coding (GSC) to efficiently code 3DGS-based scene representations [2]. Within GSC, 3DGS data are regarded as high-dimensional attribute tensors similar to point clouds, and the standardization effort is being pursued in parallel along two approaches. First, a fast-track approach is being discussed to extend and amend existing point cloud compression standards, namely Video-based Point Cloud Compression (V-PCC) and Geometry-based PCC (G-PCC) [3, 4], to make them suitable for 3DGS coding. Second, during the exploratory phase of GSC, a dedicated video-based compression framework for 3DGS—MPEG GSC Test Model Candidate Video (TMCV) [5]—has been developed and evaluated, where 3DGS attributes are organized into 2D attribute maps and then encoded using HEVC.

Among 3DGS attributes, SH coefficients account for a large portion of the data size and are therefore a key target for improving compression efficiency. However, existing frameworks provide limited tools that exploit the structural correlation among SH coefficients. For example, the video-based approach leverages spatial correlation only implicitly by feeding attribute maps into a video codec, while mechanisms that directly reduce correlation among coefficients arising from the SH indexing structure are not sufficiently considered. In addition, in V-PCC-based 3DGS coding, RGB-to-YUV color space conversion is applied as a preprocessing step to mitigate inter-channel correlation of SH coefficients; nevertheless, substantial correlation remains within each channel (intra-channel) and within the coefficient structure even after the conversion, indicating further room for redundancy reduction.

Based on these observations, this paper proposes a prediction-based residual coding method that exploits the correlation structure among SH coefficients to effectively remove redundancy in SH data. To validate the generality and practicality of the proposed method, we implement it in both the MPEG GSC TMCV framework and a V-PCC-based framework, and demonstrate compression efficiency improvements through performance evaluation under the common test conditions (CTC) [6].

2. VIDEO-BASED CODING FRAMEWORKS IN GSC

2.1 Test Model Candidate Video (TMCV)

In MPEG GSC, the MPEG GSC is used as a test model

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for video-codec-based I-3DGS compression. TMCV adopts a framework in which each attribute of a 3DGS model (e.g., position, color, rotation, scale, opacity, and SH coefficients) is quantized on an attribute-by-attribute basis and converted to an integer representation, and then organized into 2D frame-based attribute maps that are encoded with HEVC. To strengthen spatial correlation within the attribute maps, the Parallel Linear Assignment Sorting (PLAS) [7] algorithm is applied to reorder the Gaussian primitives, and each attribute map is generated according to the reordered primitive order.

The generated attribute maps are then arranged into multiple video sequences per attribute and compressed using HEVC, and the resulting HEVC bitstreams are multiplexed into a Visual Volumetric Video-based Coding (V3C) bitstream. In the decoding process, per-attribute images are reconstructed from the V3C bitstream, and the reconstructed attribute values are converted back into 3D Gaussian parameters to render free-viewpoint images.

2.2 V-PCC Amendment for GSC

MPEG is pursuing standardization activities to amend the existing G-PCC and V-PCC standards for 3DGS coding under a fast-track approach toward rapid standardization of GSC. Conventional V-PCC partitions a point cloud into patches, projects each patch onto a 2D atlas image based on geometry information, and compresses the generated atlas using a 2D video codec such as HEVC. However, since 3DGS includes multiple high-dimensional attribute tensors, it is necessary to extend the original point-cloud-oriented V-PCC pipeline to efficiently compress 3DGS models.

In this context, the GSC group is discussing a 3DGS-specific V-PCC extension based on the RAW Toolset [8]. The RAW Toolset skips the geometry-based patch generation and projection stages of conventional V-PCC, and instead directly maps the attribute values of each Gaussian primitive onto a 2D atlas while preserving a consistent indexing structure. This enables the diverse 3DGS attributes to be represented as independent attribute atlases; in the considered configuration, a total of 21 attribute atlases is generated and encoded using HEVC. In the decoding process, per-attribute images are reconstructed from the bitstream, and the reconstructed attribute values are converted back into 3D Gaussian parameters.

3. CORRELATION ANALYSIS OF SH COEFFICIENT

SH coefficients in a 3DGS model are key attributes that represent the color and directional illumination characteristics of Gaussians. SH coefficients account for a relatively large portion of the overall 3DGS attribute data and are therefore an important target for improving compression efficiency.

In this study, to analyze the correlation among SH coefficients, we compute correlation coefficients over the CTC datasets and visualize the results. As shown in the left panel of Figure 1, strong correlations are observed for many pairs of SH coefficients, indicating that redundancy exists within the SH data. This

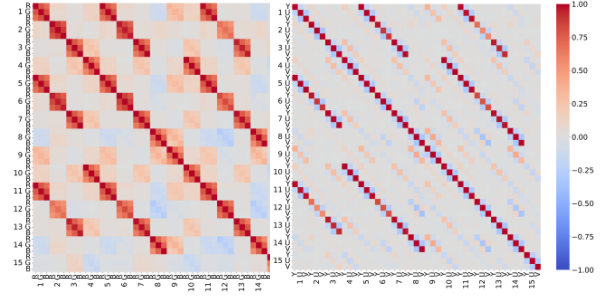


Figure 1: Correlation-coefficient heatmaps of SH coefficients before and after RGB-to-YUV color-space conversion. (Left: RGB domain, Right: YUV domain)

suggests that removing redundancy via prediction between highly correlated coefficients can improve coding efficiency.

However, although the MPEG GSC TMCV framework adopts a video-based approach in which each 3DGS attribute—including SH coefficients—is arranged into 2D attribute maps and encoded with HEVC, it does not include a dedicated tool that explicitly exploits the correlation among SH coefficients. Consequently, there are limitations in directly reducing the redundancy inherent in the SH coefficient structure, leaving room for introducing additional tools that leverage the observed correlations.

Meanwhile, in a V-PCC-based 3DGS coding framework, RGB-to-YUV color space conversion is applied as a preprocessing step to mitigate inter-channel correlation among the RGB color channels of SH coefficients. Nevertheless, even after the conversion, high correlation remains within each channel (intra-channel), as shown in the right panel of Figure 1. This implies that color space conversion alone is insufficient to fully remove redundancy in SH coefficients, and that additional correlation-reduction techniques that consider the SH coefficient structure are required.

4. PROPOSED SH COEFFICIENT PREDICTION

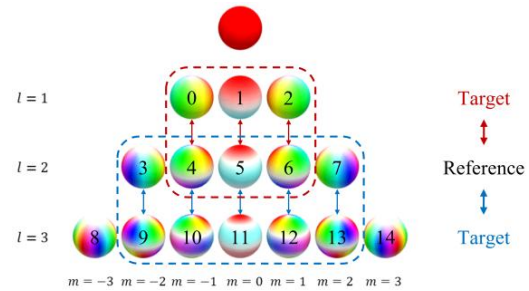


Figure 2: The proposed SH coefficient prediction.

This paper presents a residual-based coding method that exploits prediction between highly correlated SH coefficients. As indicated by the arrows in Figure 2, the proposed method leverages the correlation among structurally related SH coefficients and transmits the prediction error (residual) derived from a reference coefficient, rather than directly sending the original

target coefficient values. Since highly correlated coefficients tend to have similar values, the residual generally exhibits lower variance and lower entropy than the original signal. When residuals are mapped to 2D grid-based attribute images, the residual-based images typically become smoother and more low-frequency dominant, enabling HEVC to exploit spatial correlation more effectively and thereby improving compression efficiency. Because both TMCV and V-PCC share a common structure of mapping 3DGS attributes onto a 2D grid and then applying video coding, this statistical advantage of residual signals is applicable to both frameworks.

More specifically, in the TMCV- and V-PCC-based frameworks, 3DGS attribute values are originally represented as 32-bit floating-point numbers and are converted into integer representations via a quantization process for video coding. To mitigate error accumulation due to quantization, the proposed method performs prediction in the SH coefficient domain prior to quantization and computes residuals. The residual computation for target coefficients is defined in (1). The residual signals are then quantized and mapped onto a 2D grid. As a result, the positions corresponding to the target coefficients are filled with quantized residual values rather than the original coefficient values, and the resulting residual-based attribute images are compressed using HEVC.

$$Residual = \frac{[Target - Reference]}{2} \quad (1)$$

$$Target_{recon} = [Residual + Reference] \times 2 \quad (2)$$

At the decoder, the attribute images are decoded from the bitstream, and the residual signals are recovered through dequantization. Finally, the target coefficients are reconstructed using the recovered reference coefficients and residuals, as given in (2). Through this procedure, the target SH coefficients are restored, and the complete set of SH coefficients is subsequently reconstructed.

5. EXPERIMENTAL RESULTS

We implemented the proposed method in both TMCV [5], the video-based compression framework for GSC, and TMC2 [9], the V-PCC reference software, and evaluated its performance under the GSC CTC. Specifically, we used 15 sequences from the GSC mandatory dataset, consisting of three forward facing sequences and eleven object-centric sequences. Rendering quality was assessed using the GSC metrics [10], and BD-rate was computed using the official GSC evaluation template. Table 1 reports the coding performance in TMCV, while Table 2 reports the performance in V-PCC.

Tables 1 and 2 summarize the BD-rate performance of the proposed method relative to the TMCV anchor and the V-PCC anchor, respectively. For the forward facing sequences, the proposed method consistently outperformed the anchors in both frameworks. In the TMCV setting, the average BD-rate improved by -16.6% in terms of RGB-PSNR, with the largest gain of -21.3% observed for *breakfast_semitracked*. In the

V-PCC setting, the average BD-rate improvement was -3.5% (RGB-PSNR), and the best result of -8.5% was again achieved on *breakfast_semitracked*. In contrast, for *breakfast_untracked* in V-PCC, the BD-rate increased by +3.2% (RGB-PSNR).

Table 1: Performance of proposed method (V-PCC Anchor)

Foreward facing sequence	BD-rate RGB-PSNR	BD-rate YUV-PSNR	BD-rate YUV-SSIM
bartender_semitracked	-6.2%	-6.1%	-2.6%
cinema_semitracked	-2.5%	-2.1%	-0.6%
breakfast_semitracked	-8.5%	-8.4%	-5.6%
breakfast_untracked	3.2%	3.1%	4.8%
Average	-3.5%	-3.4%	-1.0%
Object-centric sequences	BD-rate RGB-PSNR	BD-rate YUV-PSNR	BD-rate YUV-SSIM
manwithfruit_tracked	0.4%	0.4%	0.0%
lego_ferrari	8.7%	8.8%	10.9%
lego_bugatti	8.9%	9.1%	11.6%
cricket_player	7.8%	7.4%	6.1%
plant	10.5%	10.3%	9.0%
solo_tango_female	11.3%	12.3%	9.4%
solo_tango_male	15.1%	16.3%	9.6%
tango_duo	11.4%	12.1%	11.4%
tennis_player	11.6%	12.0%	5.3%
flowerdance	8.8%	8.5%	7.8%
gymnast	13.3%	13.3%	19.9%
Average	9.8%	10.0%	9.2%

Table 2: Performance of proposed method (TMCV Anchor)

Foreward facing sequence	BD-rate RGB-PSNR	BD-rate YUV-PSNR	BD-rate YUV-SSIM
bartender_semitracked	-18.3%	-17.0%	-11.1%
cinema_semitracked	-15.0%	-13.8%	-6.0%
breakfast_semitracked	-21.3%	-20.4%	-10.6%
breakfast_untracked	-11.7%	-10.2%	-1.2%
Average	-16.6%	-15.3%	-7.2%
Object-centric sequences	BD-rate RGB-PSNR	BD-rate YUV-PSNR	BD-rate YUV-SSIM
manwithfruit_tracked	-9.6%	-8.9%	-16.6%
lego_ferrari	10.2%	10.8%	12.1%
lego_bugatti	6.7%	8.6%	10.2%
cricket_player	6.3%	7.4%	8.2%
plant	9.3%	10.3%	15.4%
solo_tango_female	11.0%	12.3%	14.1%
solo_tango_male	11.6%	13.5%	12.9%
tango_duo	12.9%	14.7%	17.7%
tennis_player	9.2%	10.5%	10.4%
flowerdance	1.3%	1.9%	1.7%
gymnast	6.5%	8.0%	13.0%
Average	6.9%	8.1%	9.0%

For the object-centric sequences, an average BD-rate increase was observed in both frameworks. In the TMCV setting, the average BD-rate increased by +6.9% (RGB-PSNR); while an improvement of -9.6% was observed for *manwithfruit_tracked*, the BD-rate increased for many sequences (e.g., the *lego/solo/tango* sequences). In the V-PCC setting, the average BD-rate also increased by +9.8% (RGB-PSNR), and *manwithfruit_tracked* showed near-neutral performance with +0.4%.

6. CONCLUSION

This paper proposed a prediction-based residual coding method that exploits correlation among SH coefficients. The proposed method was implemented in both TMCV- and V-PCC-based 3DGS compression frameworks, and its performance was evaluated under the GSC CTC. Experimental results on forward-facing sequences showed average BD-rate reductions of 16.6% relative to the TMCV anchor and 3.5% relative to the V-PCC anchor in terms of RGB-PSNR. These results demonstrate that the overall coding efficiency of 3DGS can be improved by efficiently reducing inherent redundancy through prediction among SH coefficients. In particular, the results indicate that even after applying RGB-to-YUV color space conversion in the V-PCC framework to mitigate inter-channel correlation, the remaining intra-channel correlation can be further reduced.

In contrast, for object-centric sequences, average BD-rate increases were observed in both frameworks. Therefore, future work should investigate the correlation characteristics of SH coefficients and develop methods to adaptively apply the proposed SH coefficient prediction method based on these characteristics, along with appropriate signaling of its application.

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